

Satya Venkatesh Kosaraju

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Education

M.Sc. in Agricultural and Biosystems Engineering (3.26/4) 2026

Thesis Title: Optimization of Dry Fertilizer Spreaders: DEM-Based Material Flow Analysis and Experimental Validation

Iowa State University, Ames, IA, USA

Bachelor of Technology in Agricultural Engineering (8.31/10) 2023

Vignan's Foundation for Science, Technology, and Research, Guntur, India

Professional Experience

Research Assistant 2023 - 2024

(Collaboration with Soil Machine Dynamics Laboratory, Iowa State University, Ames, IA & John Deere)

- Optimization of Dry Fertilizer Spreaders: DEM-Based Material Flow Analysis and Experimental Validation
- Developed a DEM model for dry fertilizer particles to analyze and improve distribution uniformity.
- Created CAD models and optimized equipment configurations through simulation and field validation.

Patent

Mechatronic Robotic Device for Self-Regulating and Precision Irrigation Using Robotic Operating System 2020 - 2021

Authors: Satya Venkatesh Kosaraju, Dr. Balamurugan Karnan, and Dr. T. P. Latchoumi.

- Patent No: 202141009919
(https://ipindia.gov.in/writereaddata/Portal/IPOJournal/1_4961_1/Part-1.pdf)
- Developed an autonomous robot with CNN-based weed detection and precision spraying for irrigation and targeted herbicide application.

Technical Skills

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- AutoCAD
 - SolidWorks
 - Fusion360
 - Creo
 - EDEM
 - HyperMesh

- CFD
- Python
- ImageJ
- AcuSolve
- MATLAB
- JMP data analysis

Soft Skills

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- Technical Collaboration
 - Teamwork
 - Critical thinking
 - Decision making
 - Adaptability
 - Resilience

Leadership and Service

Iowa State University Engineering Career Services, Ames, IA (Ambassador)	2023
Assisted in organizing the event, providing guidance and support to recruiters and student attendees.	
National Farm Progress Show, Boon, IA (Ambassador)	2024
Presented the soil machine dynamic laboratory work (Soil Compaction) to the farmers and companies.	
World Food Price International Conference, Des Moines, IA (Volunteer)	2024
Assisted in planning and organizing the Norman E. Borlaug International Dialogue Event 2024.	

Presentations

Center for Multiphase Flow Research and Education Conference, Ames, IA	2023
Title: Developing Digital Twins of Cleaning Shoe Separation of Corn Biomass Particles	
Authors: Satya Venkatesh Kosaraju, Jong-Myung Noh, Mehari Z. Tekeste, Ravi Teja Chaganti, Yumeng Zhao	

Undergraduate Projects

Mechatronic Robotic Device for Self-Regulating and Precision Irrigation Using Robotic Operating System	2020 - 2021
Solar Integrated Briquetting Machine	2020 - 2021
Integrated Hydroponics (Plants, Chicken, Fish)	2021 - 2021
Solar Integrated Sustainable Egg Hatching Incubator	2022 - 2023
Tractor Mounted Multi Crop Seedling Transplanter	2022 - 2023

Awards

Gold medal for the best outgoing student of the year, 1/106 students, Vignan University, Guntur, India	2023
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Professional Affiliations

- Member of the American Society of Agricultural and Biological Systems Engineering - Member of the Center for Multiphase Flow Research and Education - Member of the Food and Agriculture Organization of the United Nations	
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References

Dr. Mehari, Tekeste
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